

GOVIND UNNIKRISHNAN

Physics BS-MS Graduate

govind21@iisertvm.ac.in | IISER Thiruvananthapuram Kerala, India

EDUCATION

Integrated BS-MS (5th Year. Physics Major, Data Science Minor)

2021 – 2026

Indian Institute of Science Education and Research, Thiruvananthapuram

CGPA: 8.42

RESEARCH EXPERIENCE

Low Temperature Adiabatic Demagnetization Refrigeration (ADR) Studies, Dr. Deepshikha Jaiswal-Nagar

Synthesized ADR materials via solid-state reactions and investigated their magnetic and thermodynamic properties. Performed adiabatic demagnetization refrigeration measurements on a candidate material, achieving temperatures down to 165 mK with a holding time exceeding 12 hours, demonstrating its suitability for magnetic refrigeration. Identified two field-induced critical points at millikelvin temperatures that enhanced entropy accumulation and contributed to the ultralow temperature performance. These features were confirmed through heat capacity measurements down to 100 mK, and H-T phase diagrams were constructed to map the critical behaviour and identify the quantum critical end point.

Crystal Electric Field Driven Effective Spin-1/2 Physics in a Rare-Earth Hyperkagome Lattice, Dr. Deepshikha Jaiswal-Nagar

Investigated effective spin-1/2 behaviour arising from crystal electric field (CEF) effects in the rare-earth hyperkagome lattice $\text{Li}_3\text{Nd}_3\text{Te}_2\text{O}_{12}$. Verified the stabilization of an effective spin-1/2 ground state through magnetization, heat capacity, and adiabatic demagnetization refrigeration (ADR) measurements. Analysed the CEF splitting and its role in governing the low-temperature thermodynamic response. Analyzed inelastic neutron scattering (INS) data to investigate the detailed crystal-field scheme and magnetic excitations of the rare-earth system.

High Entropy Alloys for Room Temperature Refrigeration Applications, Dr. Deepshikha Jaiswal-Nagar

Explored intermetallic compounds in varied stoichiometries to engineer and design rare-earth-free HEAs. Simulated and identified Tc using SPR-KKR (DFT) and Monte Carlo methods. Performed synthesis using arc melting and quartz sealing units, followed by structural and magnetic characterization.

Reinforcement Learning for Timetable Optimization, Dr. Saptarshi Bej

Developed a reinforcement learning algorithm to minimize timetable clashes, achieving a zero-clash schedule and extending the model for multi-slot subject optimization. Implemented parallelized optimization strategies and constraint handling via Lagrange multiplier-based walkers to efficiently explore the solution space and identify global minima.

INTERNSHIPS AND OTHER PROJECTS

Semester Project: Predicting Material Usability Using Machine Learning

01/2025 – 04/2025

Supervisor: Dr. Chandrakala Meena

- Developed machine learning models to predict material suitability for engineering applications using mechanical-property datasets.
- Trained and evaluated Logistic Regression, Random Forest, XGBoost, SVM, and ANN models on a dataset of 1552 materials; selected the optimal model based on accuracy and F1-score.

Internship Project: Low-Temperature Transport Measurements Using a Closed-Cycle Cryostat

10/2024 – 12/2024

- Manually operated a two-stage pulse-tube closed-cycle helium cryostat, including helium precooling, flow regulation, and VTI temperature control.
- Performed temperature- and magnetic-field-dependent 2-/4-probe transport measurements and developed MATLAB automation for instrument control and data acquisition.

Control Systems Project

09/2023 – 12/2023

- Designed and built a self-balancing robot and reaction wheel system using PID control.
- Programmed microcontrollers for real-time sensor integration and closed-loop control using Python and embedded systems

Quantum Materials Research, Supervisor: Dr. Deepshikha Jaiswal Nagar

04/2023 – 10/2023

- Synthesized GdCrTiO_5 (GCTO), and $\text{Ba}_6\text{Yb}_2\text{Ti}_4\text{O}_{17}$ (BYTO) for low-temperature magnetic studies.
- Investigated the magnetic ground state and spin-glass behavior of ZnNiTiO_4 (ZNT0) spinel.
- Assisted with the design and implementation of an adiabatic demagnetization refrigeration (ADR) puck for sub-kelvin measurements.

- Studied foundational concepts in solid-state physics through standard texts, including introduction to Solid State Physics (Kittel) and Solid State Physics (Ashcroft & Mermin).

EXPERIMENTAL EXPERIENCE

Material Synthesis

Experienced in solid-state reactions, single-crystal synthesis, arc melting, and tube sealing techniques for alloy synthesis, with hands-on experience in sputtering techniques for thin-film growth.

Cryogenic instrumentation

- Transport measurements (Performed R-T (2,4 probe and van der Pauw methods), Hall effect, and thermal transport studies.)
- Magnetic and thermodynamic measurements (PPMS, MPMS)
- Ultralow temperature transport and specific heat measurements (using ^3He - ^4He dilution refrigerator)

Experimental Characterization

Powder XRD, single-crystal XRD, Rietveld refinement; ESR, XPS, Raman spectroscopy

COMPUTATIONAL AND THEORETICAL EXPERTISE

- Electronic Structure & Magnetism:** DFT calculations using SPR-KKR and Quantum ESPRESSO for geometry optimization, SCF, NSCF, band structure, DOS/PDOS. spin-polarized and magnetic property analysis, crystal-field modeling using PyCrystalField.
- Finite-Temperature & Spin Modeling:** Computation of exchange interactions via the LKAG formalism and Monte Carlo simulations to study magnetic phase transitions and estimate T_c .
- Numerical & Data Analysis:** Proficient in Python and MATLAB for numerical analysis, data fitting, thermodynamic modeling, machine learning, and reinforcement learning.
- Visualization & Post-Processing:** Experience with VESTA, Origin, GNUPlot, Tableau, GaussView and FullProf.

ACADEMIC AND EXTRACURRICULAR ACTIVITIES

- Volunteered at the Global Science Fest, demonstrating an Arduino-based project or human electromagnetic signal analysis using EOG and EMG data.
- Developed a Python script for automated certificate generation during the FSP Symposium
- Attended Frontier symposium in Physics 2025 held at IISER TVM from 17th - 19th January 2025. Frontier Symposium in Physics 2024 held at IISER TVM from 19th - 21st January 2024. UGC DAE conference 14th December 2025, Conference on Quantum Materials and Devices on 9th August 2024 etc.
- Learned violin professionally and active football player

RELEVANT COURSES

- Physics Major:** Condensed Matter Physics I & II, Quantum Mechanics I & II, Classical Mechanics, Mathematical Methods, Electronics, Semiconductor Physics, Statistical Mechanics, Electrodynamics and Special Theory of Relativity, Atomic and Molecular Physics, Computational Techniques and Programming Languages.
- Data Science Minor:** Machine Learning, Scientific Computing, Data Analysis and Visualization
- Electives:** Modelling Materials, Experimental Physics, Electrochemical Energy Systems, Machine Learning for Physical Sciences, Statistical and Data Analysis Methods in Physical Sciences, Advanced Physics Laboratories (4 undergraduate and 3 core labs).

PUBLICATIONS

Unraveling the Field-Induced Magnetism in the Frustrated Garnet $\text{Gd}_3\text{Te}_2\text{Li}_3\text{O}_{12}$ with Enhanced Adiabatic Demagnetization Refrigeration

Dheeraj Ranaut*, Govind Unnikrishnan, Samraj M., Sharath Kumar C., Prashanta K. Mukharjee, Philipp Gegenwart and Deepshikha Jaiswal-Nagar (Manuscript under review (accepted for publication in "ACS Applied Electronic Materials"))

REFERENCES

Dr. Deepshikha Jaiswal-Nagar, Associate Professor, IISER Thiruvananthapuram

deepshikha@iisertvm.ac.in

Dr. Saptarshi Bej, Assistant Professor, IISER Thiruvananthapuram

sbej7042@iisertvm.ac.in